

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	1
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	1
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

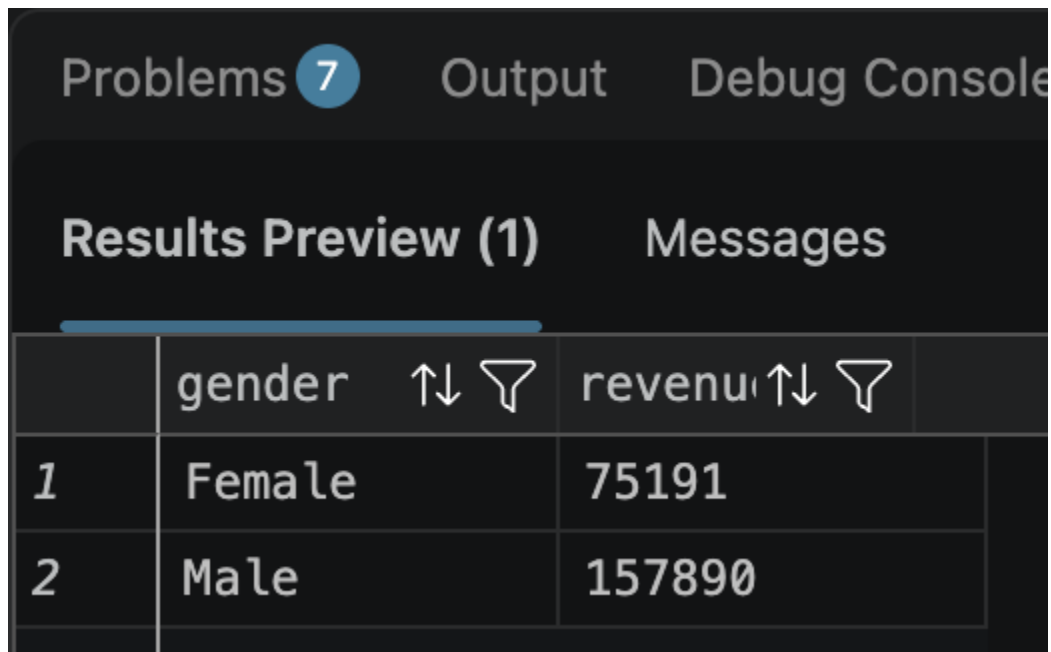
Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to AzureSQLEdge run on Docker and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in AzureSQLEdge to answer key business questions:

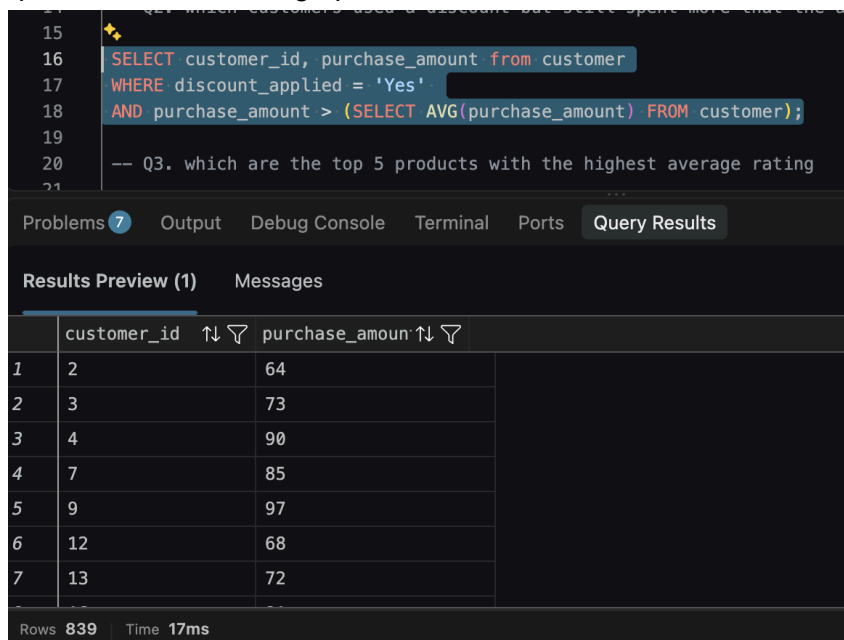
1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.



The screenshot shows the SQL Edge interface with the 'Results Preview (1)' tab selected. The table displays the following data:

	gender	revenue
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.



The screenshot shows the SQL Edge interface with a query executed. The query is:

```
SELECT customer_id, purchase_amount from customer
WHERE discount_applied = 'Yes'
AND purchase_amount > (SELECT AVG(purchase_amount) FROM customer);
```

The 'Query Results' tab is selected, showing the following data:

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72

Rows: 839 | Time: 17ms

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

```
22 SELECT TOP 5
23 item_purchased, ROUND(AVG(review_rating), 2) AS avg_rating
24 FROM customer
25 GROUP BY item_purchased
26 ORDER BY AVG(review_rating) DESC;
27
28 -- Q4. What is the average purchase amount between standard and
```

Problems 7 Output Debug Console Terminal Ports Query Results

Results Preview (1) Messages

	item_purchased	avg_rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

```
29
30 SELECT shipping_type, AVG(purchase_amount) AS avg_purchase
31 FROM customer
32 GROUP BY shipping_type
33 HAVING shipping_type IN ('Standard', 'Express');
34
35 -- Q5. Do subscribed customers spend more than non-subscribed
```

Problems 7 Output Debug Console Terminal Ports Query Results

Results Preview (1) Messages

	shipping_type	avg_purchase
1	Express	60
2	Standard	58

Page 2 Time: 10ms

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

```
36
37 SELECT subscription_status,
38 COUNT(customer_id) AS customer_count,
39 AVG(purchase_amount) AS avg_purchase,
40 SUM(purchase_amount) AS total_revenue
41 FROM customer
42 GROUP BY subscription_status
43 ORDER BY total_revenue, avg_purchase DESC;
44
45 -- Q6. Which 5 products have the highest percentage of purchases with the discount
46
```

Problems 7 Output Debug Console Terminal Ports Query Results

Results Preview (1) Messages Open

	subscription_status	customer_count	avg_purchase	total_revenue
1	Yes	1053	59	62645
2	No	2847	59	170436

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

```
47 SELECT TOP 5 item_purchased,
48 ROUND((SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) * 100.0) / COUNT(*), 2) AS discount_percentage
49 FROM customer
50 GROUP BY item_purchased
51 ORDER BY discount_percentage DESC;
52
53 -- 07. Segments customers into New, Returning, and Loyal
```

Problems 7 Output Debug Console Terminal Ports Query Results

Results Preview (1) Messages Open in New Tab

	item_purchased	discount_percentage
1	Hat	50.0000000000
2	Sneakers	49.6600000000
3	Coat	49.0700000000
4	Sweater	48.1700000000
5	Pants	47.3700000000

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

```
54 -- based on their purchase history and show the count for each segment
55 with customer_type as (
56     SELECT customer_id, previous_purchases,
57     CASE
58         WHEN previous_purchases = 1 THEN 'New'
59         WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
60         ELSE 'Loyal'
61     END AS customer_type
62     FROM customer
63 )
64
65 SELECT customer_type, COUNT(customer_id) AS customer_count
66 FROM customer_type
67 GROUP BY customer_type
68 ORDER BY customer_count DESC;
69
70 -- 8. What are the top 3 most purchased items in each category?
```

Problems 7 Output Debug Console Terminal Ports Query Results

Results Preview (1) Messages

	customer_type	customer_count
1	Loyal	3116
2	Returning	701
3	New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each Category.

```
71
72 WITH item_counts AS (
73     SELECT category, item_purchased, COUNT(customer_id) AS purchase_count,
74     ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS rank
75     FROM customer
76     GROUP BY category, item_purchased
77 )
78
79 SELECT rank, category, item_purchased, purchase_count
80 FROM item_counts
81 WHERE rank <= 3
82 ORDER BY category, rank;
83
```

Problems 7 Output Debug Console Terminal Ports Query Results

Results Preview (1) Messages Open in N

	rank	category	item_purchased	purchase_count
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

```
86 SELECT subscription_status,
87        COUNT(customer_id) AS repeat_buyers
88 FROM customer
89 WHERE previous_purchases > 5
90 GROUP BY subscription_status
91
92 -- Q10. What is the revenue contribution of each age group?
```

Problems 7 Output Debug Console Terminal Ports

Results Preview (1) Messages

	subscription_status	repeat_buyer
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

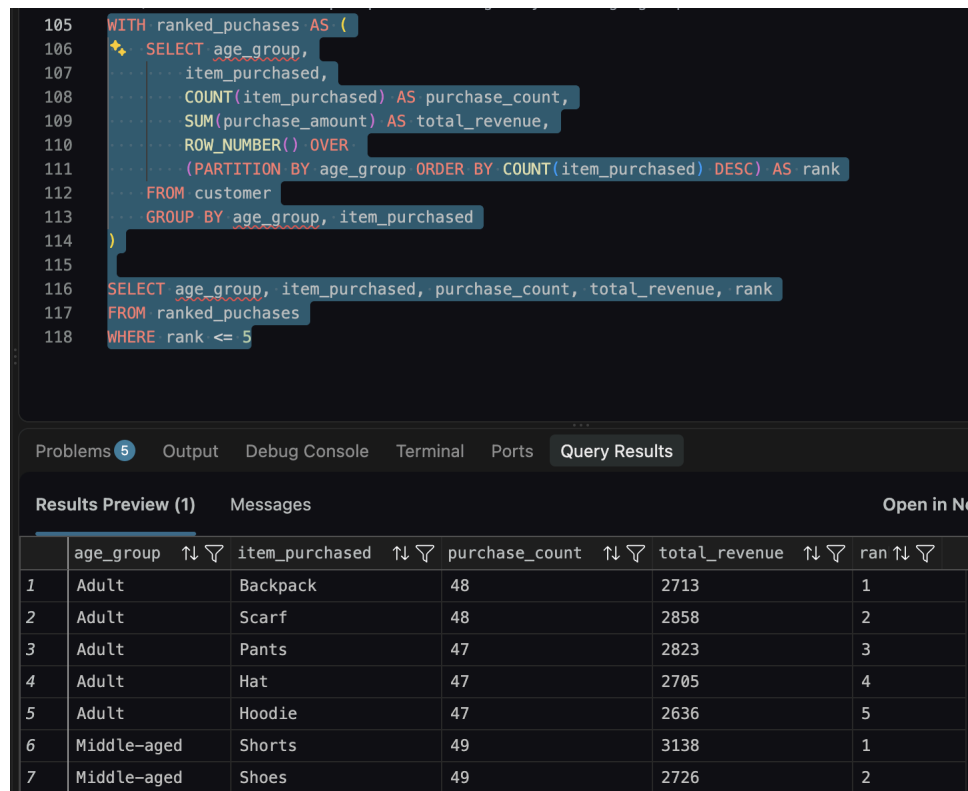
```
94 SELECT age_group,
95        SUM(purchase_amount) AS total_revenue,
96        COUNT(customer_id) AS customer_count,
97        ROUND(CAST((SUM(purchase_amount) * 100.0) / (SELECT SUM(purchase_amount)
98        FROM customer) AS DECIMAL(10,2)), 2) AS revenue_percentage
99 FROM customer
100 GROUP BY age_group
101 ORDER BY total_revenue DESC;
102
103 -- Q11. Which age group that has the subscription are the most profitable customers?
```

Problems 7 Output Debug Console Terminal Ports Query Results

Results Preview (1) Messages

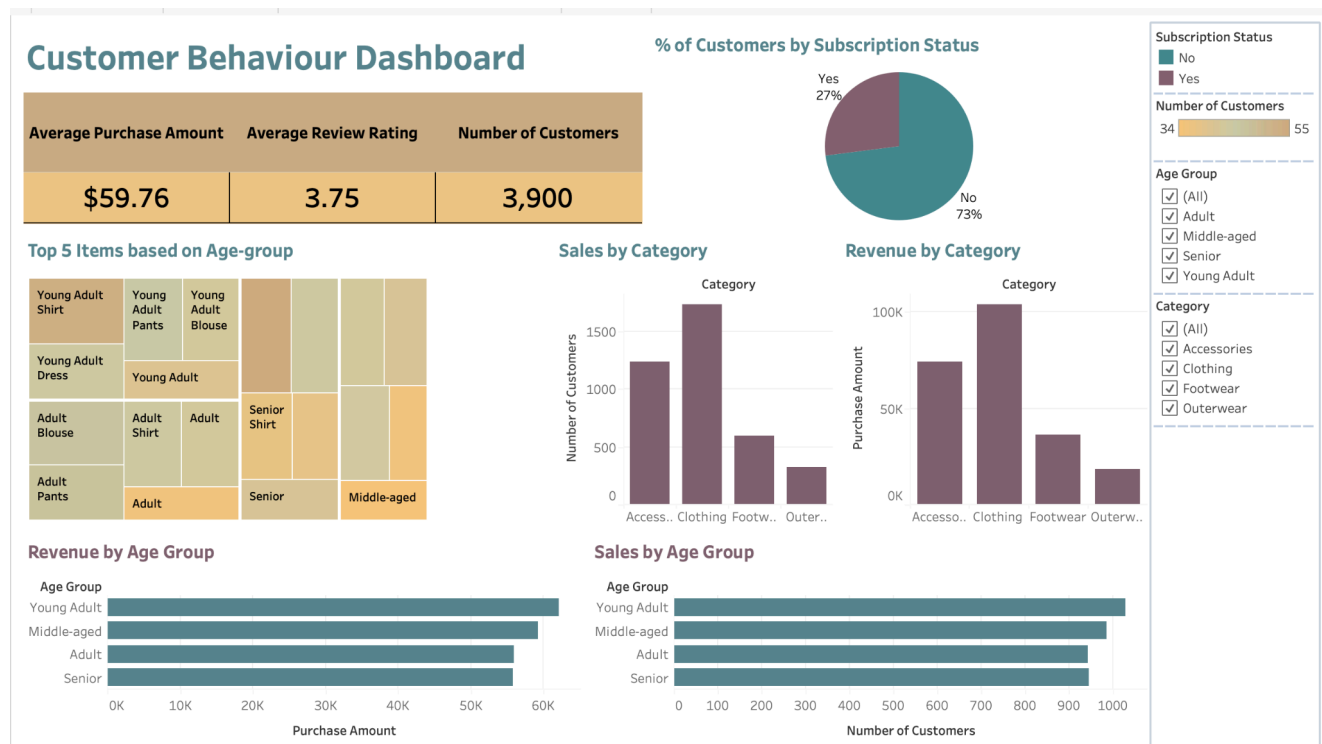
	age_group	total_revenue	customer_count	revenue_percentage
1	Young Adult	62143	1028	26.66
2	Middle-aged	59197	986	25.40
3	Adult	55978	942	24.02
4	Senior	55763	944	23.92

11. **Top 5 Products Bought by each Age-group**– Ranked the top 5 products bought by each Age-group and how much revenue it generated.



5. Dashboard in Tableau

Finally, we built an interactive dashboard in **Tableau** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers to encourage more customers to sign up. Offer discounts on selected bags and clothing products to attract young adults, and provide gifts or discounts to senior customers who spend more on jewellery.
- **Review Discount Policy** – Evaluate and adjust the current discount strategy to increase sales while maintaining healthy profit margins.
- **Improve Product Positioning** – Highlight top-rated and best-selling products in marketing campaigns to increase product visibility and encourage customer purchases.
- **Implement Targeted Marketing** – Focus marketing efforts on high-revenue age groups and customers who use express shipping, as these segments demonstrate stronger purchasing behaviour and revenue potential.